

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

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**Flipped class room activity report on
Design of Traffic Light System using Arduino Uno R3 in
TinkerCAD simulation tool**

**DEPARTMENT
OF
ELECTRONICS AND COMMUNICATION ENGINEERING**

Submitted by

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S.J.M VIDYAPEETHA ®

S.J.M INSTITUTE OF TECHNOLOGY

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

(Affiliated to Visvesvaraya Technological University, Belagavi, Recognized by AICTE,
New Delhi and Approved by Government of Karnataka)

NAAC Accredited with 'B⁺⁺' Grade

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1. Introduction

Traffic lights are essential control systems used to manage vehicle movement at road intersections and pedestrian crossings. They help reduce accidents and maintain smooth traffic flow by displaying standardized signals such as **Red (Stop)**, **Yellow (Ready/Wait)**, and **Green (Go)**. In this project, a **Traffic Light Control System** is designed and simulated using **Arduino Uno R3** in the **TinkerCAD virtual designing tool**. This simulation provides practical understanding of microcontroller-based control systems without the need for physical hardware.

2. Objectives

- To study the working principle of a traffic light system
- To design and simulate a traffic light controller using Arduino Uno R3
- To learn interfacing of LEDs with Arduino digital pins
- To understand timing control using delay functions
- To observe sequential operation of traffic signals in a virtual environment

3. Software Requirements

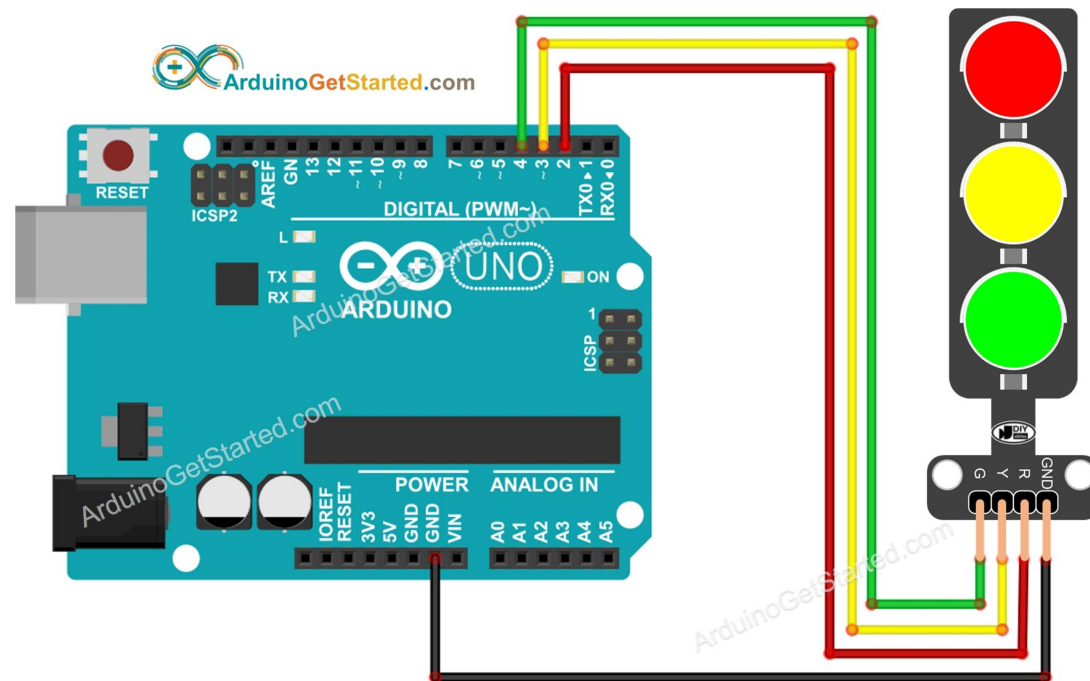
- TinkerCAD (Online Virtual Simulation Tool)
- Arduino Uno R3 (Virtual Board)
- Arduino IDE (Built-in Code Editor in TinkerCAD)
- Web Browser (Chrome / Firefox / Edge)
- Active Internet Connection

4. Steps to Access TinkerCAD

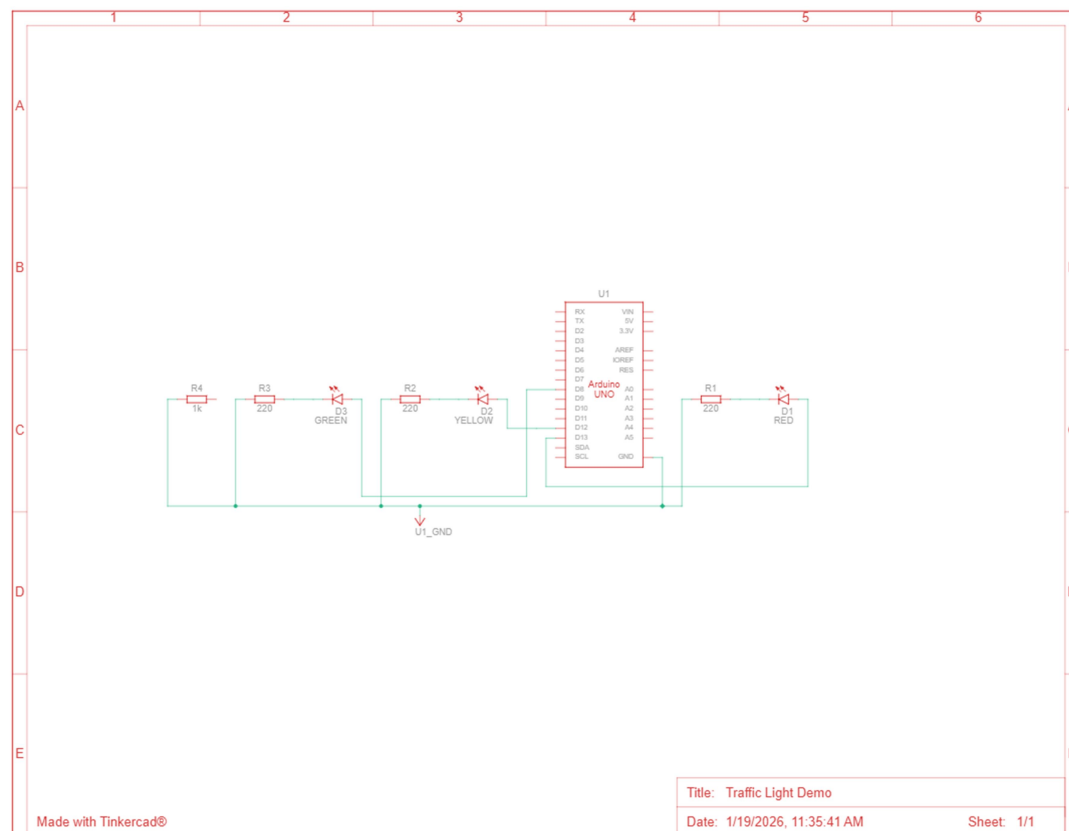
1. Open a web browser and visit **www.tinkercad.com**
2. Log in using a Google account or create a new account
3. From the dashboard, click on **Circuits**
4. Select **Create New Circuit**
5. Add **Arduino Uno R3**, LEDs, and resistors from the components panel
6. Connect the circuit according to the traffic light configuration
7. Open the **Code Editor** and write the Arduino program
8. Click on **Start Simulation** to execute the project

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Circuit Diagram

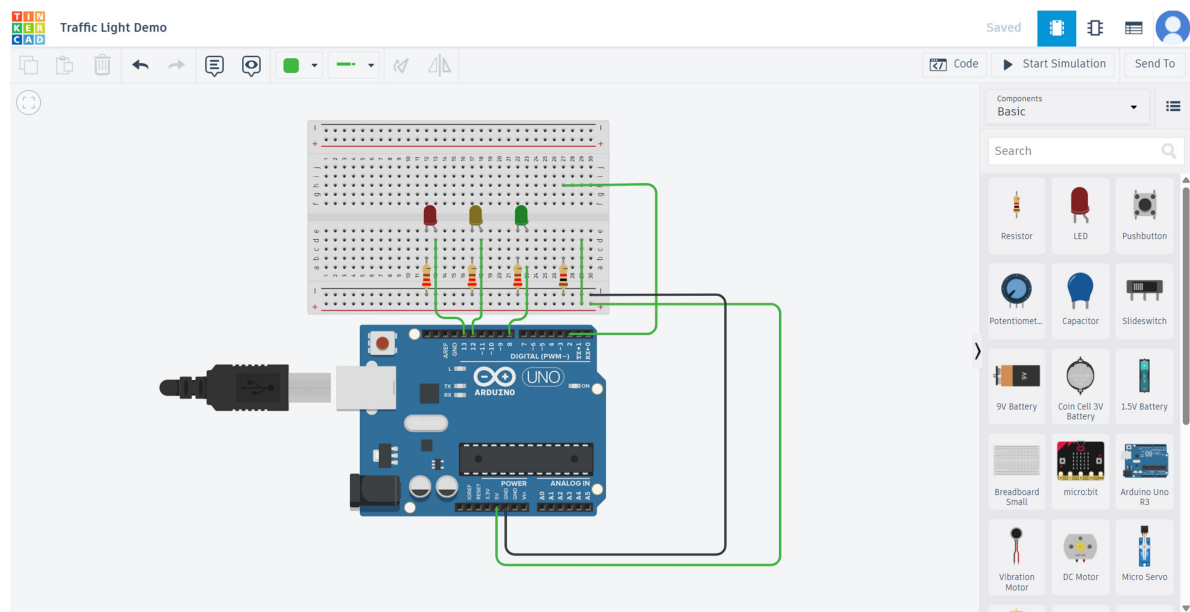


Schematic Diagram



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Circuit View

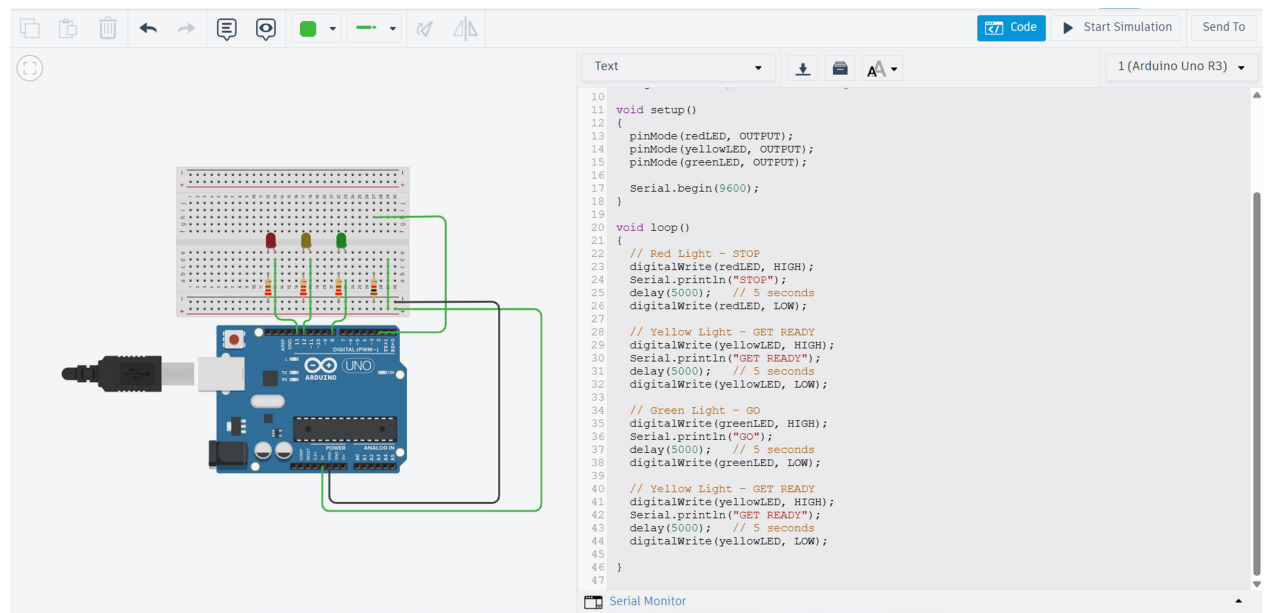


Components Required

Component List			Download CSV
Name	Quantity	Component	
D1	1	Red LED	
D2	1	Yellow LED	
D3	1	Green LED	
R1 R2 R3	3	220 Ω Resistor	
R4	1	1 k Ω Resistor	
U1	1	Arduino Uno R3	

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Code Required to Simulation



Required Code in Text

*/*Street Traffic Light Control Program*

Red, Yellow, and Green LEDs glow sequentially

Each LED stays ON for 5 seconds/*

```
int redLED = 13;    // Red light
int yellowLED = 12; // Yellow light
int greenLED = 8;   // Green light

void setup()
{
  pinMode(redLED, OUTPUT);
  pinMode(yellowLED, OUTPUT);
  pinMode(greenLED, OUTPUT);

  Serial.begin(9600);
}

void loop()
```

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```
{  
  
  // Red Light - STOP  
  
  digitalWrite(redLED, HIGH);  
  
  Serial.println("STOP");  
  
  delay(5000); // 5 seconds  
  
  digitalWrite(redLED, LOW);  
  
  
  // Yellow Light - GET READY  
  
  digitalWrite(yellowLED, HIGH);  
  
  Serial.println("GET READY");  
  
  delay(5000); // 5 seconds  
  
  digitalWrite(yellowLED, LOW);  
  
  
  // Green Light - GO  
  
  digitalWrite(greenLED, HIGH);  
  
  Serial.println("GO");  
  
  delay(5000); // 5 seconds  
  
  digitalWrite(greenLED, LOW);  
  
  
  // Yellow Light - GET READY  
  
  digitalWrite(yellowLED, HIGH);  
  
  Serial.println("GET READY");  
  
  delay(5000); // 5 seconds  
  
  digitalWrite(yellowLED, LOW);  
  
}
```

5. Observation

- The **Red LED** glows first, indicating the STOP signal
- After a specific delay, the **Yellow LED** turns ON, indicating GET READY
- The **Green LED** then glows, indicating GO
- Each LED remains ON for a fixed duration
- The sequence repeats continuously, simulating real traffic lights

5. Advantages of Traffic Light System

- ✓ **Enhances Road Safety**
Traffic lights reduce the risk of accidents by controlling vehicle and pedestrian movement at intersections.
- ✓ **Prevents Traffic Congestion**
Systematic control of traffic flow helps avoid traffic jams and confusion.
- ✓ **Ensures Smooth Traffic Flow**
Vehicles move in an orderly manner, improving overall road efficiency.
- ✓ **Reduces Human Dependency**
Automated signals reduce the need for traffic police at busy junctions.
- ✓ **Minimizes Human Error**
Fixed signal timing reduces mistakes caused by manual traffic control.
- ✓ **Improves Pedestrian Safety**
Traffic lights provide safe crossing time for pedestrians.
- ✓ **Saves Time and Fuel**
Proper traffic regulation reduces unnecessary waiting and fuel consumption.
- ✓ **Cost-Effective in Long Term**
Once installed, traffic lights require minimal maintenance.
- ✓ **Works in All Weather Conditions**
Traffic lights operate reliably regardless of weather conditions.
- ✓ **Supports Modern Smart Traffic Systems**
Can be integrated with sensors, cameras, and IoT for intelligent traffic management.

6. Result and Analysis

- ✓ The Traffic Light System using Arduino Uno R3 was successfully designed and simulated in TinkerCAD.
- ✓ The Red, Yellow, and Green LEDs operated in the correct sequence as per the programmed logic.
- ✓ The Red LED remained ON for the maximum duration, ensuring proper stop time for vehicles.
- ✓ The Yellow LED functioned as a transition signal between Red and Green states.
- ✓ The Green LED allowed smooth vehicle movement for a fixed time interval.
- ✓ Arduino Uno R3 accurately controlled the digital output pins without any errors.

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- ✓ The delay function provided stable and consistent timing for each traffic signal.
- ✓ Serial Monitor output confirmed correct execution of the program logic.
- ✓ No signal overlap was observed, indicating proper timing synchronization.
- ✓ The system continuously repeated the traffic light cycle without interruption.
- ✓ The virtual simulation helped verify circuit connections and code correctness.
- ✓ The project demonstrates the effectiveness of microcontroller-based traffic control systems.
- ✓ The design can be extended for real-time applications by adding sensors or pedestrian signals.

7. Conclusion

The Traffic Light Control System using Arduino Uno R3 was successfully designed and simulated using the TinkerCAD virtual designing tool. The project demonstrated the basic working principle of traffic signals by controlling Red, Yellow, and Green LEDs in a predefined sequence with accurate timing. This simulation helped in understanding the role of microcontrollers in automation and real-time control systems, as well as the importance of timing and logic in traffic management.

Overall, the project provided practical knowledge of Arduino programming, digital output control, and circuit simulation without the need for physical hardware. The Traffic Light System proved to be reliable, efficient, and easy to implement in a virtual environment. The project can be further enhanced by integrating sensors, pedestrian signals, or adaptive timing methods, making it suitable for advanced and real-world traffic control applications.